

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA4301

COURSE NAME: Internet Programming

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Assessment Tool : Data Interpretation

1. HTTP Response Analysis

Given Header:

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: -150

Connection: keep-alive

1. Incorrect / Invalid Fields

Field	Status	Reason
HTTP/1.1 200 OK	Valid	Correct HTTP version and success status.
Date	Valid	Correct date format.
Content-Type	Valid	Valid MIME type.
Content-Length: -150	Invalid	Length cannot be negative.
Connection: keep-alive	Valid	Persistent connection supported.

2. Impact of Invalid Content-Length

- Content-Length specifies the number of bytes in the response body.
- Negative values violate HTTP standards.
- Browsers or proxies may reject the response, truncate content, or close the connection unexpectedly.

3. Status Code Analysis

Status code 200 OK indicates the request was processed successfully. However, because the response contains an invalid header, the overall response is malformed. The body and headers must both be valid for a proper 200 OK response.

4. Correct Header

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: 150

Connection: keep-alive

2. DOM Structure Analysis

Original Code:

```
<html><head><title>Test Page</title></head><body><div><p>Hello  
World</div></body></html>
```

Question	Answer
Missing Elements	Closing <code></p></code> tag is missing.
Browser Interpretation	Browser automatically inserts the missing closing tag while building the DOM.
Impact	Rendering is usually correct, but invalid HTML may affect CSS styling and JavaScript DOM traversal.
Correction	Close the paragraph before closing the div.

Correct Code:

```
<html>  
<head><title>Test Page</title></head>  
<body>  
<div>  
<p>Hello World</p>  
</div>  
</body>  
</html>
```

3. GET vs POST in Login System

Feature	GET	POST
Data Location	URL	Request Body
Visible	Yes	No
Bookmarkable	Yes	No
Suitable for Login	No	Yes
Security	Low	Moderate (with HTTPS)

GET exposes usernames and passwords in browser history, logs, bookmarks, and shared links. POST hides credentials from the URL, and when combined with HTTPS, secure password hashing, secure cookies, CSRF protection, and MFA, provides a much safer authentication mechanism.

4. HTML Code Inspection

Problems Found:

- Missing `</h1>`
- Missing `</p>`
- `<img>` lacks alt attribute.

Correct Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome</h1>
<p>This is a paragraph</p>

</body>
</html>
```

5. Browser–Server Communication Flow

Step	Explanation
1. URL	User enters website address.
2. DNS Lookup	Domain name converted to IP address.
3. HTTP Request	Browser sends request to web server.
4. Server Processing	Server processes request and prepares response.
5. HTTP Response	Server sends HTML/CSS/JS and other resources.
6. Rendering	Browser builds DOM/CSSOM and displays webpage.

Possible Delays: DNS lookup, slow network, server processing, database queries, large files, JavaScript execution.

Performance Optimizations:

- DNS caching
- Browser caching
- Content Delivery Network (CDN)
- Gzip/Brotli compression
- Image optimization
- Minified CSS/JS
- HTTP/2 or HTTP/3